

Pre-Load Data Quality Checklist

Bank Loan Analytics Project

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Purpose: Before a new Excel/CSV file is uploaded to Snowflake, these checks help catch important data-quality issues at the source. The checklist focuses on checks that are not already handled by the current Snowflake pipeline.

Rule: Never silently fix questionable business data. If the correct value can be verified from an authoritative source, correct it. Otherwise, flag or reject the record. Do not guess.

1. Checks Required Before Upload

Check	Instruction
ID	ID must not be NULL or blank, and IDs must be unique. Investigate duplicates before upload.
Negative numeric values	Check financial/numeric fields for negative or clearly invalid values. Investigate; correct only from an authoritative source or flag/reject it.
Invalid dates	Check that required dates are valid and consistently formatted. Do not invent a date.
Loan status	Expected values include Charged Off, Fully Paid, Current . Never guess or fill a status without investigation.
Other unexpected categories	Check important category fields for unexpected values. Verify before changing them.
Missing mandatory values	Check fields that are required by the business process. Do not replace missing values with guesses or zero unless approved.
Numeric format	Make sure numeric fields contain valid numbers and no obvious text/error values.
File structure	Make sure required columns are present, names match the agreed RAW table, and there are no accidental extra rows or columns.

2. What the Snowflake Pipeline Already Handles

Already handled	Current pipeline behavior
Leading/trailing spaces	TRIM() standardizes accidental spaces.
Blank text values	NULLIF() converts blank text to NULL where used.
Missing EMP_TITLE	COALESCE() represents missing EMP_TITLE as Unknown .
Incoming duplicate IDs	ROW_NUMBER() keeps one incoming record per ID before the MERGE.
Existing vs new IDs	MERGE updates a matching ID and inserts a new ID.

3. Simple Examples

Negative numeric value: Never change **-50,000** → **50,000** just because it looks wrong. Investigate; correct only from an authoritative source or flag/reject it.

Loan status: If the value should be **Charged Off, Fully Paid, or Current**, never guess or fill it without investigation.

Missing value: Do not fill a missing business value with a guessed value. Use an approved rule or flag it.

4. Final Upload Check

■ ID is present and unique ■ Negative/invalid numeric values checked ■ Dates checked ■ Loan status/categories checked
■ Required fields checked ■ Numeric format checked ■ File structure checked

5. Responsibility

Source team: performs these basic checks before upload. **Snowflake:** handles the downstream cleaning and incremental processing already implemented in the Stream/Task pipeline.

Reference: Snowflake and Microsoft Learn documentation was used as technical reference for the relevant platform features.